Problem

Title

Harshad (Niven) Number Check

Topics

Number Mathematics, Digit Extraction, Divisibility Rules, Loops, Conditional Statements

Difficulty Level

Medium

Problem Detailed Statement (Scenario Based)

In a number system workshop, students are introduced to **Harshad Numbers**, also known as **Niven Numbers**.

A **Harshad Number** is defined as a number that is divisible by the sum of its digits.

In simple terms:

1. Extract all digits of the number.
2. Compute the sum of those digits.
3. Check whether the original number is divisible by that sum.
4. If it is divisible, it is called a Harshad Number.

For example:

- 18
  Digits → 1 + 8 = 9
  18 ÷ 9 = 2 ✔
- 21
  Digits → 2 + 1 = 3
  21 ÷ 3 = 7 ✔

However:

- 19
  Digits → 1 + 9 = 10
  19 ÷ 10 ≠ integer ✖

Now, the trainer provides a positive integer **N**. Students must determine whether the given number is a Harshad Number.

The implementation must:

- Use loop logic to extract digits (% and / operators).
- Avoid string conversion.
- Use conditional statements to check divisibility.
- Handle edge cases carefully (especially when digit sum becomes 0).

This problem improves:

- Digit manipulation skills
- Divisibility reasoning
- Conditional logic application
- Loop control

Edge cases to consider:

- N = 0
- Single digit numbers
- Large input values

Input Format

A single integer:

`N`

Constraints:

`0 ≤ N ≤ 10^9`

Output Format

- Print `"Harshad Number"` if N is a Harshad Number.
- Otherwise print `"Not Harshad Number"`.

Examples

Example 1

Input:

```
18
```

Output:

```
Harshad Number
```

Example 2

Input:

```
21
```

Output:

```
Harshad Number
```

Example 3

Input:

```
19
```

Output:

```
Not Harshad Number
```

Example 4

Input:

```
1729
```

Output:

```
Harshad Number
```

(1 + 7 + 2 + 9 = 19, and 1729 ÷ 19 = 91)